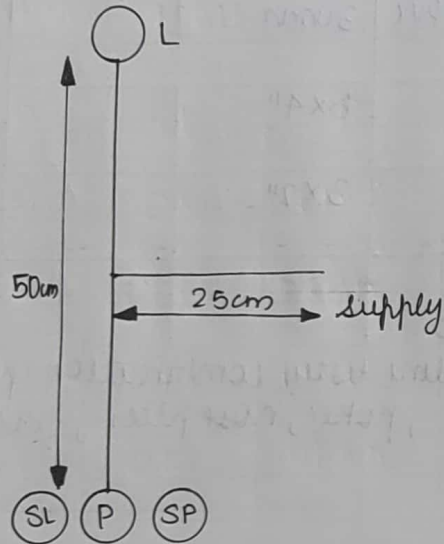
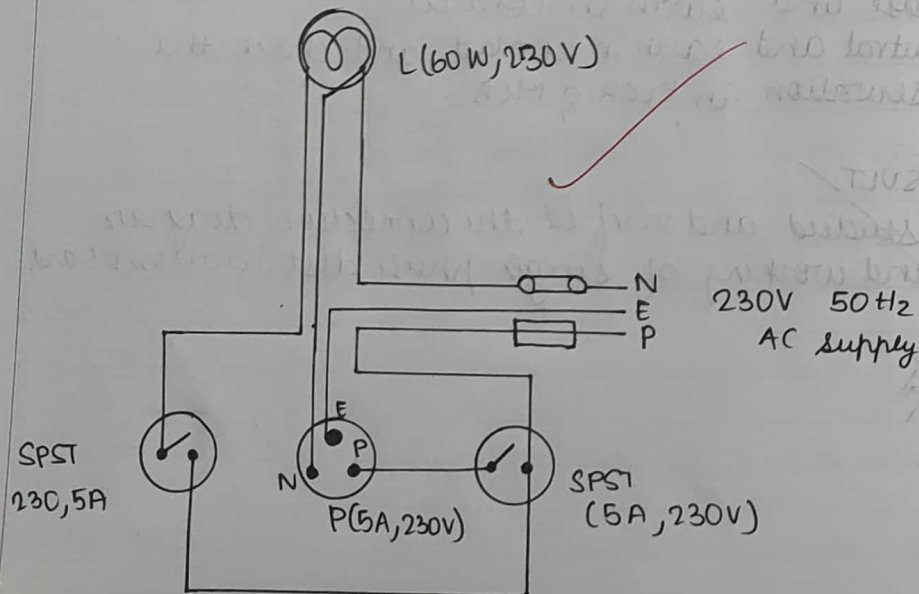


LAYOUT DIAGRAM



CIRCUIT DIAGRAM



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PLUG POINT WIRING AND FLUORESCENT TUBE

A) PLUG POINT WIRING

AIM

wire up a circuit to obtain one plug and one lamp is controlled by independent switches

DESIGN

$$\text{Power, } P = VI \cos \phi$$

$$P = 500W + 60W$$

(500W → Plug load)
(60W → Lamp load)

$$\text{Load current} = \frac{P}{230} = \frac{560}{230} = \underline{\underline{2.4347A}}$$

$$\begin{aligned} \text{Fuse Wire Rating} &= \text{load current} \times 1.5A \\ &= 2.4347 \times 1.5A \\ &= \underline{\underline{3.6521A}} \end{aligned}$$

OBSERVATION TABLE

SP	SL	P	L
ON	ON	LIVE	BRIGHT
OFF	OFF	DEAD	DARK
ON	OFF	LIVE	DARK
OFF	ON	DEAD	BRIGHT

NOTE

All dimensions are in cm

L → lamp

P → plug

SP → switch for plug

SL → switch for lamp

F → Fuse

NL → Neutral link

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MATERIALS REQUIRED

S.No	Items	Specification	Rating	Quantity
1)	Switch	SPST	5A, 230V	2
2)	Lamp	Incandescent	60W, 250V	1
3)	Conduit	20mm PVC		75cm
4)	Copper wire	1 sq mm		250cm
5)	Lamp holder	Straight batten	5A	1
6)	Junction	20mm PVC 3 way		1
7)	Round Block	3 inch PVC		1
8)	Switch box	3x2" PVC 6x4" PVC		2 1
9)	Clamp / saddle	20mm metal		3
10)	Fuse unit	Kit Kat (ordinary rewirable)	16A	1
11)	Fuse wire	Tin and lead alloy	2A	10cm

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12.	Neutral link	Porcelain bar with copper segment	16A	1
13.	Insulation tape			1
14	Plug Point socket	3 Pin	5A, 230V	1
15.	Screws	1/2"		6
		3/4"		8
		1 1/4"		1

TOOLS REQUIRED

Screw driver, line tester, combination pliers, ball-peen hammer, poker, nose plier, wire stripper

PROCEDURE

- 1) Draw the circuit diagram and check whether it is correct.
- 2) Fix the conduit to the board using saddle of appropriate diameter.
- 3) Wire the circuit as per the circuit diagram.
- 4) Fix the switch box and give its connections.
- 5) Fix the round block to the base board.

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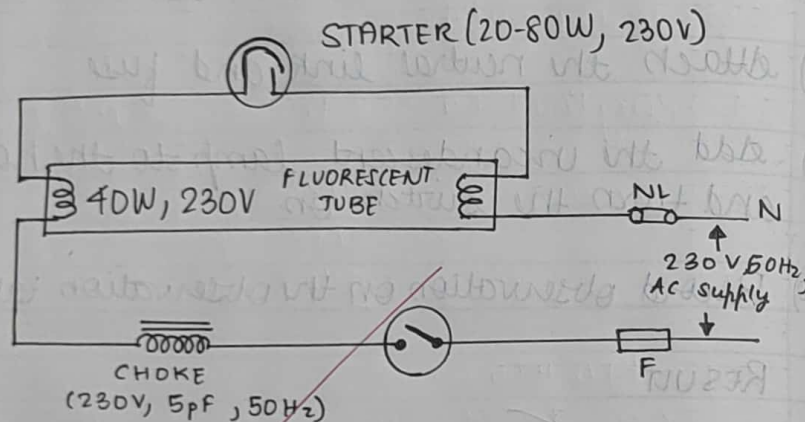
- 6) Fix the lamp holder on top of the round block and give its connection.
- 7) Fix the socket on the 6x4" switch box
- 8) Attach the neutral link and fuse
- 9) Add the incandescent lamp to the holder and turn the switch on.
- 10) Record observation on the observation table.

RESULT

Wired up the circuit and obtained one plug and one lamp controlled by independent switches.

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CIRCUIT DIAGRAM



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B) FLUORESCENT TUBE

AIM

Study the connection and working of fluorescent tube and prepare estimation for the same.

DESIGN

Power, $P = VI \cos \phi$

$$40 = 230 \times I \times 0.8$$

$$\text{Load Current, } I = \frac{40}{230 \times 0.8} = \underline{\underline{0.21739A}}$$

$$\begin{aligned} \text{Fuse wire rating} &= 1.5 \times \text{load current} \\ &= \underline{\underline{0.3260A}} \end{aligned}$$

THEORY

These lamps are hot cathode low pressure mercury vapour lamps and are manufactured in the form of long glass tube having different wattages and lengths as 20W and 0.6m (2 feet), 40W and 1.2m (4 feet), 65W, 80W and 1.5m (5 feet). There are two tungsten filaments at each end of the fluorescent tube. These are coated with electron emitting material like barium oxide for easy emission of a electron when heated. The tube is coated from inside with

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fluorescent powder and is filled with argon gas at low pressure and some mercury. The gas is added in the tube because at the time of starting the mercury is in the form of normal liquid state, so the lamp starts as conducting with argon gas. When the temperature rises, the liquid mercury changes into vapour form and takes over the conduction of current. A choke coil and a starter is also connected in series with the tube. The function of the choke coil is to provide high, impulse voltage of about 1000 volts for starting the tube and after starting it acts as a ballast in the circuit to drop the voltage. A glow type starter in the circuit performs the function of a switch.

The connection of a tube with a glow type starter and a choke coil is shown in the circuit diagram. The glow type starter has two bimetallic electrodes enclosed in a small glass tube containing helium gas. These electrodes normally remain open. When the switch is put on, a small current flows from one ~~electrode~~ electrode to the other through the helium gas.

To minimize the radio interference due to make break sound of starter and to increase the

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life of bimetallic electrodes and a small condenser is connected across the starter.

When the switch is put on, a small current flows through the choke coil, tube filaments and starter. This small current passes from one electrode to the other electrode through the helium gas in the starter. Discharge through the helium gas develops some heat which bends the bi-metallic electrode and thus they come direct in contact with each other. Now sufficiently high current flows directly through the electrodes. After one or two seconds these electrodes cool down. This results in breaking the contact and thus current stops flowing through the filament as well as choke coil. Due to this sudden break, a high voltage of about 4 to 5 times of normal voltage is induced in the choke coil which is applied on the long gas column and thus the discharge starts direct in the tube. After the starting of the tube, the choke coil works as a ballast and ~~the~~ drops voltage across it (due to increased reactance or impedance after starting) and thus the voltage across the tube is less than the supply voltage.

This lamp gives glareless cool day-light and has efficiency of 40 lumens per watt. The average life

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of the lamp is about 1000 working hours. This is useful for workshops; drawing rooms, offices and for all local and general lighting. When this lamp is required to use on ac supply, a resistance is connected in series with the choke which acts as a ballast in the beginning and later on it drops the voltage and smoothen the current in the circuit.

Fluorescent Materials. The fluorescent material used inside the tube in the form of coating absorbs the ultraviolet rays and change them into light rays. Coating of different materials give different colours of light and some of them are given below.

No	Fluorescent Material	Colour
1.	Calcium tungstate	Blue
2.	Magnesium tungstate	Blue-white
3.	Zinc Beryllium silicate	Yellow-white
4.	Cadmium silicate	Yellow-Pink
5.	Cadmium borate	Pink
6.	Zinc silicate	Green

OBSERVATION TABLE

S	1
ON	BRIGHT
OFF	DARK

NOTE:-

L → Lamp

S → Switch

F → Fuse

N → Neutral Link

SPST → Single pole single throw

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MATERIALS REQUIRED

S.No	Items	Specification	Rating	Quantity
1.	starter		20-80W, 230 V	1
2.	Fluorescent tube		40W, 230V	1
3.	Choke		230V, 5pF, 50Hz	1
4.	Switch	SPST	5A, 230V	1
5.	Fuse Unit	Kit Kat (ordinary rewirable)	16A	1
6.	Fuse Wire	Tin and lead alloy	2A	10cm
7.	Neutral link	Porcelain Bar with copper segment	16A	1
8.	Insulation tape			1
9.	Fluorescent tube holder			1

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TOOLS REQUIRED

screw drivers, line tester, combination pliers, ball pen hammer, pokes, nose pliers, wire strippers.

PROCEDURE

- 1) Draw the circuit diagram and check whether it is correct.
- 2) Wire the circuit as per the diagram
- 3) Fix the switch box and give the connection
- 4) Fix the tube holder on the board and give its connections
- 5) Attach the neutral link and fuse
- 6) Add the fluorescent tube to the holder and turn the switch on.
- 7) Record the observation on the table

RESULT

Studied the connection and working of fluorescent tube prepared estimation for the same.